

OBJECT-ORIENTED PROGRAMMING (100457-11001)
PROJECT BRIEF

ASSESSMENT DETAILS	
TITLE/NAME	GROUP PROJECT
WEIGHT	20%
DATE/DEADLINE	Please refer below
INSTRUCTIONS	Please refer below
DELIVERABLES	Written Report (Final Report) Presentation

GROUP PROJECT: OBJECT-ORIENTED PROGRAMMING

Objectives:

1. Apply core OOP concepts: classes, objects, encapsulation, inheritance, and polymorphism.
2. Use abstract classes and/or interfaces appropriately.
3. Implement exception handling and file I/O.
4. Design a clear object-oriented architecture.
5. Develop a basic GUI using an OOP approach.
6. Work collaboratively and manage responsibilities in a small development team.

Project Overview:

This project aims to provide students with hands-on experience in object-oriented design and implementation. Working in groups of four, students will design, implement, and demonstrate a medium-scale object-oriented software application that applies core OOP principles learned in the course.

Project Theme (Choose ONE)

Each group must select one application domain from the list below (or propose an equivalent theme):

- Student Management System
- Library Management System
- Online Shopping / Order Management System
- Hotel or Room Booking System
- Clinic or Appointment Scheduling System
- OOP-Based Game or Simulation (with GUI)

The system must be of moderate complexity and demonstrate meaningful use of OOP concepts.

Technical Requirements

1. Object-Oriented Design

- Minimum 8 - 10 well-designed classes.
- Proper use of:
 - Encapsulation (private fields, public methods).
 - Inheritance (at least one inheritance hierarchy).
 - Polymorphism (overridden methods or interface-based behavior).

2. Abstract Classes / Interfaces

- At least one abstract class or interface.
- Polymorphism must be demonstrated at runtime.

3. Exception Handling

- Appropriate use of try-catch (or equivalent)
- Must handle invalid user input, file errors, or logical errors
- Custom exception classes are recommended, but optional

4. File Input / Output (I/O)

- Persistent storage using files (e.g., text or serialized objects)
- Examples:
 - Saving user records
 - Loading application data at startup
 - Writing logs

5. Graphical User Interface

- All user interaction must be through a GUI.
- Supported technologies (choose one):
 - Tkinter, PyQt, or equivalent

Project Deliverables:

1. Source Code:

- Clean, well-organized project structure.
- Readable class and method names.
- Meaningful comments where needed.
- Must compile and run without errors.

2. Presentation: A short group presentation demonstrating your system.

3. Final Report: The report must include:

- Introduction & problem description.
- System features and screenshots of the GUI.
- Explanation of OOP concepts used.
- Description of GUI design and event handling.
- Team roles and individual contributions.
- Conclusion.

Assessment Guidelines:

The project will be assessed based on the following criteria:

Tasks	Marks
1. Functionality: a. Completeness of basic features. b. Correctness of the program execution, absence of bugs.	40
2. Code Quality: a. Code readability, organization, and documentation. b. Adherence to coding standards.	40
3. Project Report a. Clear and concise project report.	10
4. Presentation a. Well-organized and engaging presentation. b. Demonstration of teamwork and individual contributions.	10
Total Marks	100

Group Formation

Students will be assigned to groups consisting of **four (4)** members. Each member is expected to **contribute and participate actively** throughout the entire project process. Sharing ideas and providing assistance in completing projects among group members is required.

Important Deliverables Timeline

Item Submission	Deadline	
Group Formation	Week10	6th. May 2026
Final Report	Week 14	5th. June 2026
Project Presentation	Week 15	10th. June 2026